STAT 565 - Homework 6

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In [95]:
```
import warnings
warnings.filterwarnings('ignore')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from pathlib import Path
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.stats.diagnostic import acorr_ljungbox

plt.style.use('seaborn-v0_8-whitegrid')
```

Problem 1

In [96]:
```
data_path = Path('dpi.dat')
dpi = pd.read_csv(data_path, sep=r'\s+', header=None, names=['year', 'month'
dpi['date'] = pd.to_datetime(dict(year=dpi.year, month=dpi.month, day=dpi.da
dpi = dpi.sort_values('date').set_index('date')

x = dpi['dpi'].astype(float)
logx = np.log(x)
y = logx.diff().dropna()

print(f'Observations in x_t: {len(x)}')
print(f'Observations in y_t = diff(log x_t): {len(y)}')
dpi.head()
```

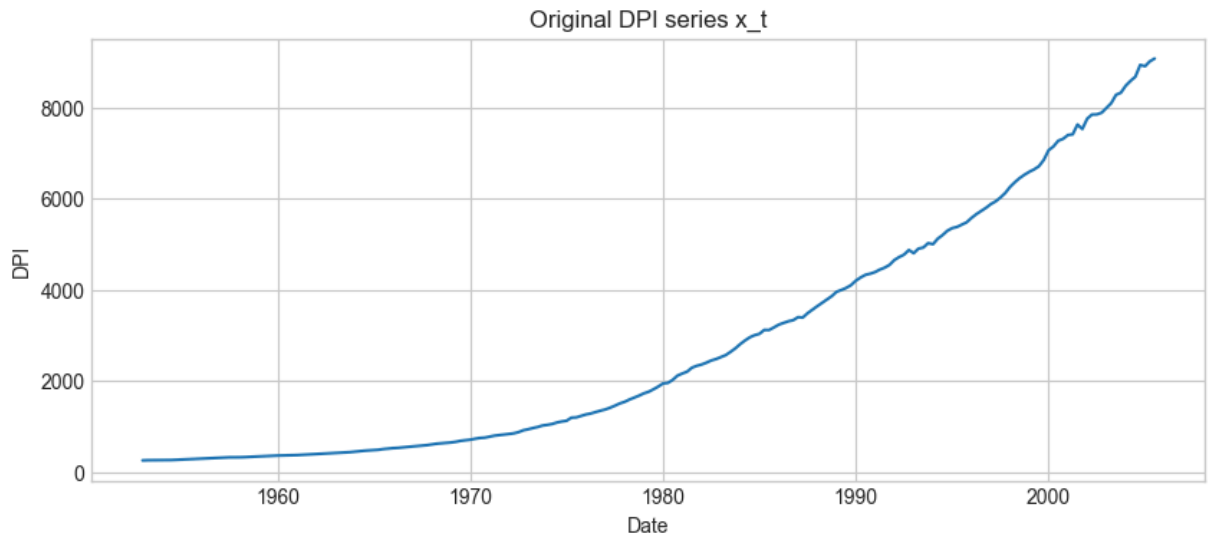
```
Observations in x_t: 211
Observations in y_t = diff(log x_t): 210
```

Out[96]:

	year	month	day	dpi
date				
1953-01-01	1953	1	1	255.6
1953-04-01	1953	4	1	259.6
1953-07-01	1953	7	1	259.9
1953-10-01	1953	10	1	259.4
1954-01-01	1954	1	1	262.2

Part 1

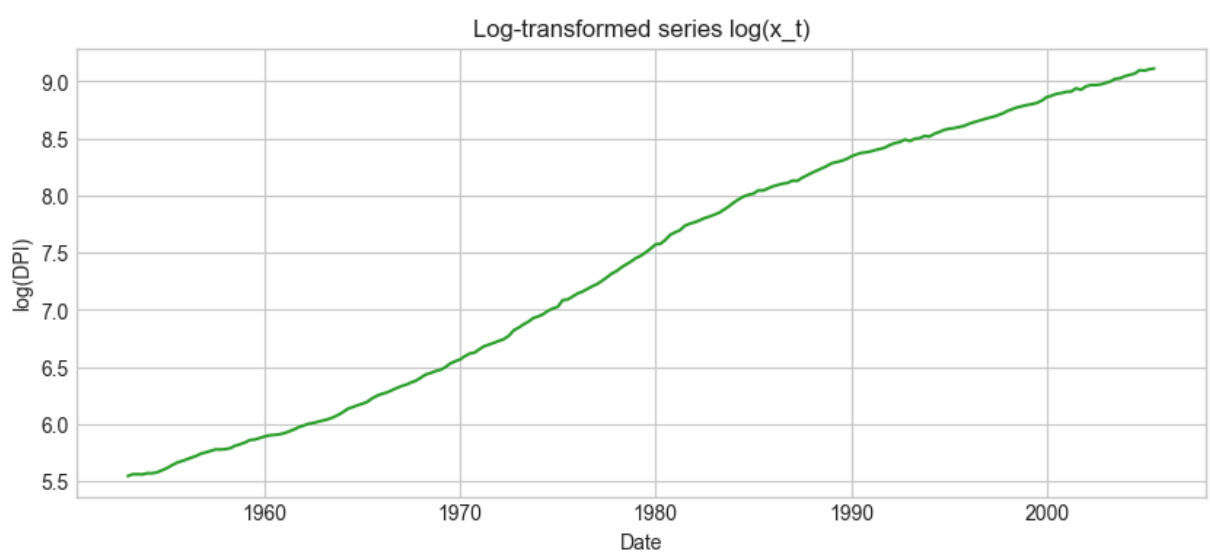
In [97]:
```
fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(x.index, x.values, color='tab:blue')
ax.set_title('Original DPI series x_t')
ax.set_xlabel('Date')
ax.set_ylabel('DPI')
plt.show()
```



x_t shows strong trend and level growth, so it is not stationary.

Part 2

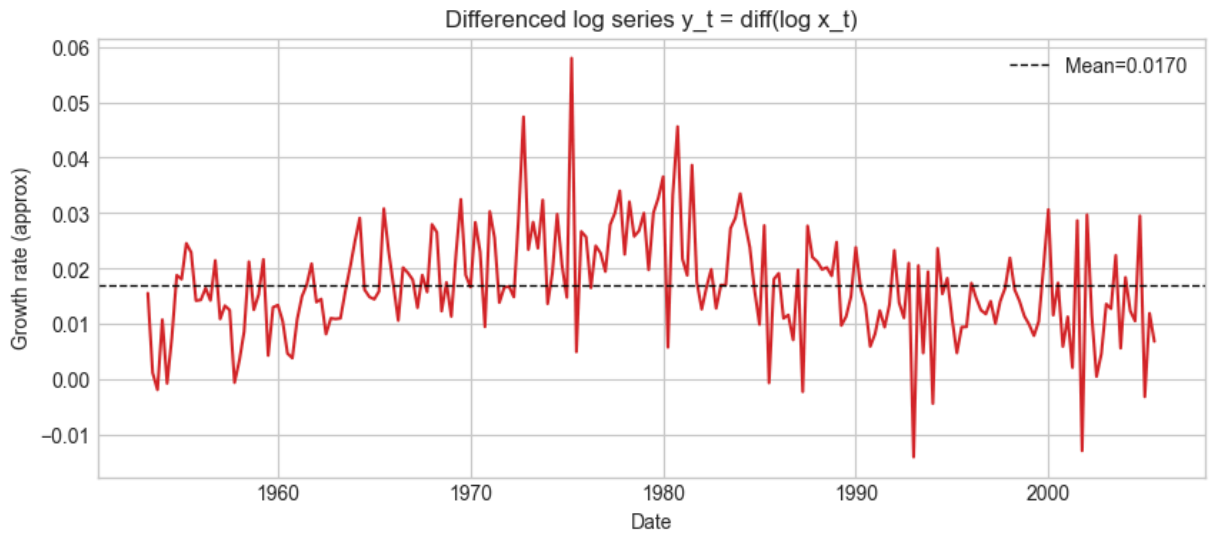
In [98]:
```
fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(logx.index, logx.values, color='tab:green')
ax.set_title('Log-transformed series log(x_t)')
ax.set_xlabel('Date')
ax.set_ylabel('log(DPI)')
plt.show()
```



log transform stabilizes variance somewhat, but trend remains -> still nonstationary.

Part 3

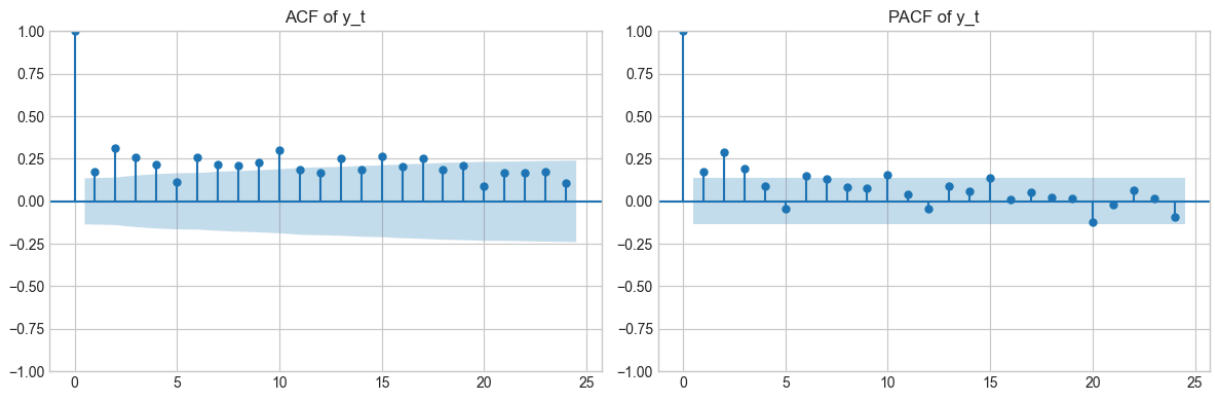
In [99]:
```
fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(y.index, y.values, color='tab:red')
ax.axhline(y.mean(), color='black', linestyle='--', linewidth=1, label=f'Mea
ax.set_title('Differenced log series y_t = diff(log x_t)')
ax.set_xlabel('Date')
ax.set_ylabel('Growth rate (approx)')
ax.legend()
plt.show()
```



y_t fluctuates around a roughly constant mean and variance; this appears close to stationary.

Part 4

In [100…
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
plot_acf(y, lags=24, ax=axes[0])
plot_pacf(y, lags=24, ax=axes[1], method='ywm')
axes[0].set_title('ACF of y_t')
axes[1].set_title('PACF of y_t')
plt.tight_layout()
plt.show()
```



In [101…

```
def eacf_like(series, max_ar=5, max_ma=5, acf_lags=24):
    from statsmodels.tsa.stattools import acf

    n = len(series)
    crit = 1.96 / np.sqrt(n)
    table = pd.DataFrame(index=[f'AR({p})' for p in range(max_ar + 1)],
                         columns=[f'MA({q})' for q in range(max_ma + 1)])

    for p in range(max_ar + 1):
        if p == 0:
            resid = series - np.mean(series)
        else:
            try:
                m = ARIMA(series, order=(p, 0, 0), trend='n').fit()
                resid = m.resid
            except Exception:
                table.loc[f'AR({p})', :] = 'NA'
                continue

        r = acf(resid, nlags=max(acf_lags, max_ma + 1), fft=False)
        for q in range(max_ma + 1):
            val = r[q + 1]
            table.loc[f'AR({p})', f'MA({q})'] = 'o' if abs(val) <= crit else

    return table

eacf_tbl = eacf_like(y, max_ar=5, max_ma=5)
print('EACF-like table (o = near-zero, x = significant):')
display(eacf_tbl)

tentative_models = [(1, 0, 1), (2, 0, 1), (1, 0, 2)]
print('Tentative models:', [f'ARMA({m[0]},{m[2]})' for m in tentative_models
```

```
EACF-like table (o = near-zero, x = significant):
```

	MA(0)	MA(1)	MA(2)	MA(3)	MA(4)	MA(5)
AR(0)	x	x	x	x	o	x
AR(1)	x	o	o	o	x	o
AR(2)	x	x	o	o	x	o
AR(3)	o	o	x	o	x	o
AR(4)	o	o	o	x	x	o
AR(5)	o	o	o	x	x	o

```
Tentative models: ['ARMA(1,1)', 'ARMA(2,1)', 'ARMA(1,2)']
```

Part 5

In [102…

```
def rolling_one_step_mse(series, order, last_n=30):
    preds, actual = [], []
    n = len(series)
    for t in range(n - last_n, n):
        train = series.iloc[:t]
        target = series.iloc[t]
        try:
            fit = ARIMA(train, order=order, trend='n').fit()
            fc = fit.forecast(steps=1).iloc[0]
            preds.append(fc)
            actual.append(target)
        except Exception:
            preds.append(np.nan)
            actual.append(target)

    preds = np.array(preds, dtype=float)
    actual = np.array(actual, dtype=float)
    mask = np.isfinite(preds)
    return np.mean((actual[mask] - preds[mask]) ** 2)

results = {}
for order in tentative_models:
    name = f'ARMA({order[0]},{order[2]})'
    fit = ARIMA(y, order=order, trend='n').fit()

    table = pd.DataFrame({'estimate': fit.params, 'std_error': fit.bse})
    resid = fit.resid
    p, q = order[0], order[2]
    lb = acorr_ljungbox(resid, lags=[12], model_df=(p + q), return_df=True)
    mse30 = rolling_one_step_mse(y, order, last_n=30)

    results[name] = {'fit': fit, 'coef_table': table, 'resid': resid, 'ljung

summary_rows = []
for name, r in results.items():
    summary_rows.append({'model': name, 'AIC': r['aic'], 'LjungBox_p_lag12':

model_summary = pd.DataFrame(summary_rows).sort_values('AIC').reset_index(dr
model_summary
```

Out[102…

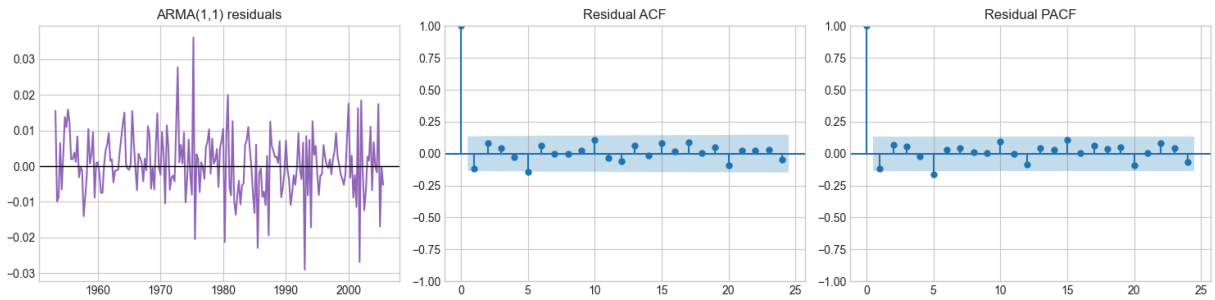
	model	AIC	LjungBox_p_lag12	Rolling_MSE_last30
0	ARMA(2,1)	-1384.287842	0.243631	0.000097
1	ARMA(1,1)	-1383.361525	0.171681	0.000102
2	ARMA(1,2)	-1377.400952	0.092576	0.000098

In [103…

```
for name, r in results.items():
    print('=' * 80)
    print(name)
    print('- Coefficients and standard errors')
    display(r['coef_table'])

    resid = r['resid']
    fig, axes = plt.subplots(1, 3, figsize=(15, 3.8))
    axes[0].plot(resid.index, resid.values, color='tab:purple')
    axes[0].axhline(0, color='black', linewidth=1)
    axes[0].set_title(f'{name} residuals')

    plot_acf(resid, lags=24, ax=axes[1])
    axes[1].set_title('Residual ACF')

    plot_pacf(resid, lags=24, ax=axes[2], method='ywm')
    axes[2].set_title('Residual PACF')

    plt.tight_layout()
    plt.show()

    print('- Residual EACF-like table')
    display(eacf_like(resid.dropna(), max_ar=4, max_ma=4))

    print('- Box-Ljung test (lag 12, adjusted df = 12 - (p+q))')
    display(r['ljung_box'])

    print(f"- AIC: {r['aic']:.3f}")
    print(f"- Rolling one-step forecast MSE (last 30): {r['mse30']:.6f}")
```

```
================================================================================
ARMA(1,1)
- Coefficients and standard errors
```

	estimate	std_error
ar.L1	0.997781	0.003227
ma.L1	-0.899826	0.033390
sigma2	0.000078	0.000006




```
- Residual EACF-like table
```

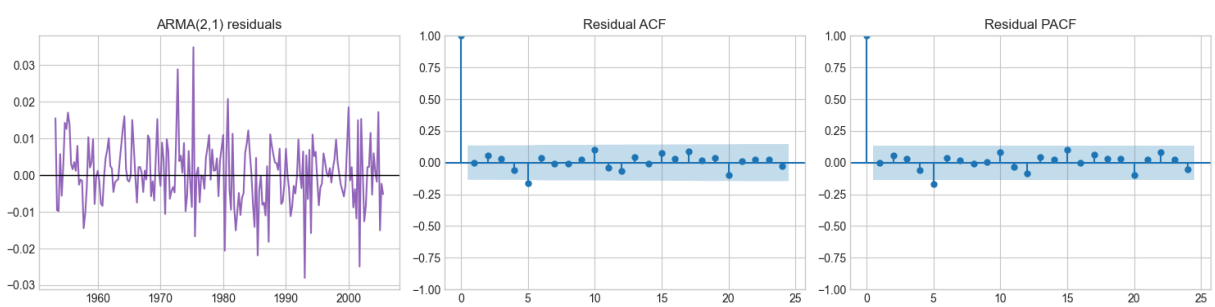
	MA(0)	MA(1)	MA(2)	MA(3)	MA(4)
AR(0)	o	o	o	o	x
AR(1)	o	o	o	o	x
AR(2)	o	o	o	o	x
AR(3)	o	o	o	o	x
AR(4)	o	o	o	o	x

```
- Box-Ljung test (lag 12, adjusted df = 12 - (p+q))
```

	lb_stat	lb_pvalue
12	14.028825	0.171681

```
- AIC: -1383.362
- Rolling one-step forecast MSE (last 30): 0.000102
================================================================================
====
ARMA(2,1)
- Coefficients and standard errors
```

	estimate	std_error
ar.L1	0.868702	0.069905
ar.L2	0.128547	0.070141
ma.L1	-0.878537	0.043839
sigma2	0.000076	0.000006




```
- Residual EACF-like table
```

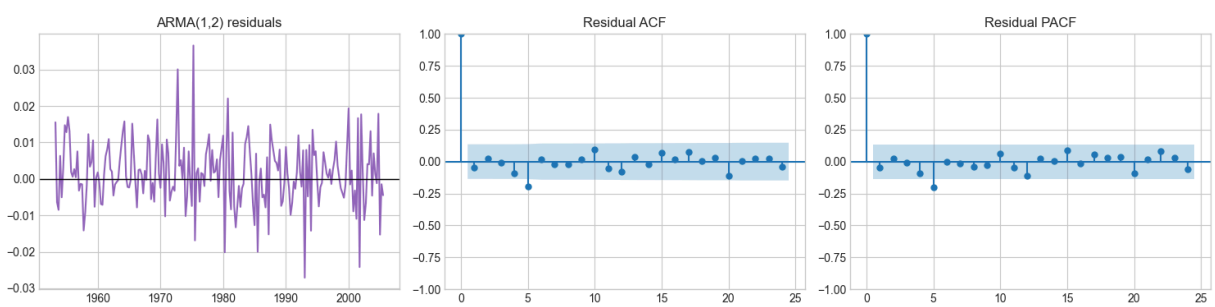
	MA(0)	MA(1)	MA(2)	MA(3)	MA(4)
AR(0)	o	o	o	o	x
AR(1)	o	o	o	o	x
AR(2)	o	o	o	o	x
AR(3)	o	o	o	o	x
AR(4)	o	o	o	o	x

```
- Box-Ljung test (lag 12, adjusted df = 12 - (p+q))
```

	lb_stat	lb_pvalue
12	11.489664	0.243631

```
- AIC: -1384.288
- Rolling one-step forecast MSE (last 30): 0.000097
==========================================================================
====
ARMA(1,2)
- Coefficients and standard errors
```

	estimate	std_error
ar.L1	0.984888	0.007985
ma.L1	-0.940920	0.066874
ma.L2	0.118547	0.067508
sigma2	0.000079	0.000006




```
- Residual EACF-like table
```

	MA(0)	MA(1)	MA(2)	MA(3)	MA(4)
AR(0)	o	o	o	o	x
AR(1)	o	o	o	o	x
AR(2)	o	o	o	o	x
AR(3)	o	o	o	o	x
AR(4)	o	o	o	o	x

```
- Box-Ljung test (lag 12, adjusted df = 12 - (p+q))
```

	lb_stat	lb_pvalue
12	14.940748	0.092576

```
- AIC: -1377.401
- Rolling one-step forecast MSE (last 30): 0.000098
```

Part 6

In [104…

```
best_by_aic = model_summary.iloc[0]['model']

print('Model comparison table:')
display(model_summary)

print(f"Selected best model: {best_by_aic}")
```

```
Model comparison table:
```

	model	AIC	LjungBox_p_lag12	Rolling_MSE_last30
0	ARMA(2,1)	-1384.287842	0.243631	0.000097
1	ARMA(1,1)	-1383.361525	0.171681	0.000102
2	ARMA(1,2)	-1377.400952	0.092576	0.000098

```
Selected best model: ARMA(2,1)
```

ARMA(2,1) has the lowest AIC

Part 7

Disposable personal income is a really interesting macroeconomic indicator linked to household consumption and overall macroeconomic conditions. In this analysis, the original DPI level series and log-transformed series show trend and are not stationary, so we model the differenced log series $y_t = \Delta(\log x_t)$. After evaluating multiple ARMA candidates using ACF/PACF/EACF-style diagnostics, residual checks, Ljung-Box tests, AIC, and rolling one-step-ahead forecast MSE over the last 30 observations, it seems that the preferred model is ARMA(2,1), which had the best AIC among entertained models and reasonable residual diagnostics for short-term forecasting

Problem 2

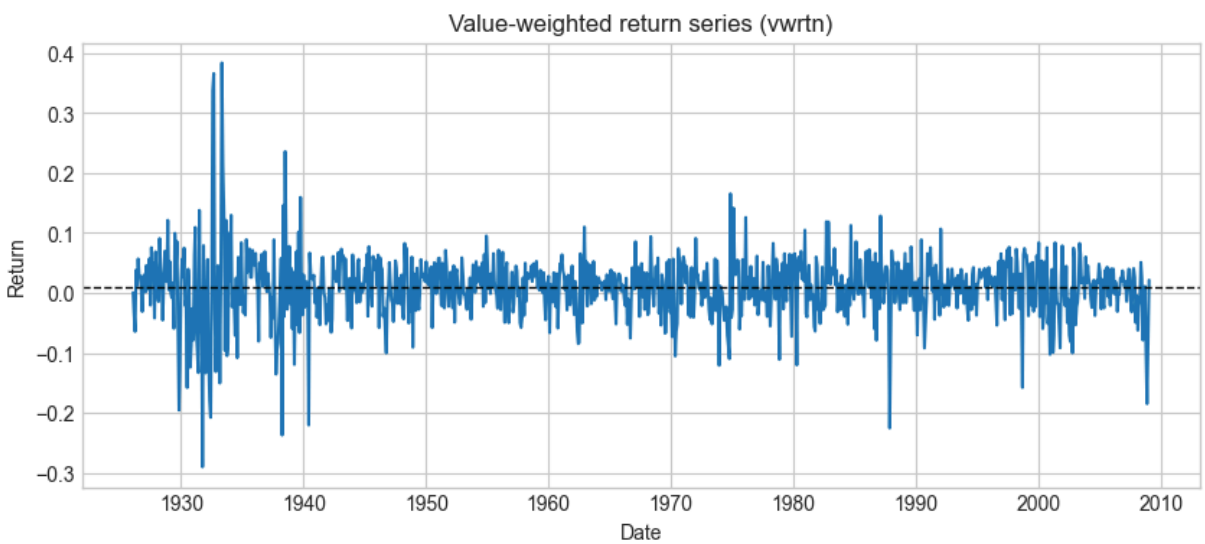
In [105…

```
vw_path = Path('m-ibm3dx2608.txt')
vw = pd.read_csv(vw_path, sep=r'\s+')
vw['date'] = pd.to_datetime(vw['date'].astype(str), format='%Y%m%d')
vw = vw.sort_values('date').set_index('date')
vw_series = vw['vwrtn'].astype(float).dropna()

print(f'Value-weighted observations: {len(vw_series)}')
display(vw.head())

fig, ax = plt.subplots(figsize=(10,4))
```

```
ax.plot(vw_series.index, vw_series.values, color='tab:blue')
ax.axhline(vw_series.mean(), color='black', linestyle='--', linewidth=1)
ax.set_title('Value-weighted return series (vwrtn)')
ax.set_xlabel('Date')
ax.set_ylabel('Return')
plt.show()
```

```
Value-weighted observations: 996
```

	ibmrtn	vwrtn	ewrtn	sprtn
date				
1926-01-30	-0.010381	0.000724	0.023174	0.022472
1926-02-27	-0.024476	-0.033374	-0.053510	-0.043956
1926-03-31	-0.115591	-0.064341	-0.096824	-0.059113
1926-04-30	0.089783	0.038358	0.032946	0.022688
1926-05-28	0.036932	0.012172	0.001035	0.007679



In [106…

```
from statsmodels.tsa.statespace.sarimax import SARIMAX

def fit_with_fixed_zeros(series, p, q, fixed_params=None):
    fixed_params = fixed_params or {}
    model = SARIMAX(series, order=(p, 0, q), trend='n',
                    enforce_stationarity=False, enforce_invertibility=False)
    if fixed_params:
        with model.fix_params(fixed_params):
            res = model.fit(disp=False)
    else:
        res = model.fit(disp=False)
    return res

def next_least_significant(res, fixed_params):
    free = []
    for name, val in res.pvalues.items():
        if (name.startswith('ar.L') or name.startswith('ma.L')) and (name no
            free.append((name, float(val)))
```

```
    if not free:
        return None
    free.sort(key=lambda z: z[1], reverse=True)
    return free[0]

def simplify_by_zeroing(series, p, q, max_steps=20):
    fixed = {}
    history = []

    for step in range(max_steps):
        try:
            res = fit_with_fixed_zeros(series, p, q, fixed)
        except Exception:
            break

        lb = acorr_ljungbox(res.resid, lags=[12], model_df=(p + q - len(fixe
        history.append({
            'step': step,
            'fixed_count': len(fixed),
            'fixed': dict(fixed),
            'aic': float(res.aic),
            'lb_pvalue_lag12': float(lb['lb_pvalue'].iloc[0]),
            'result': res
        })

        nxt = next_least_significant(res, fixed)
        if nxt is None:
            break

        name, pval = nxt
        if pval < 0.05:
            break

        fixed[name] = 0.0

    return history

def history_table(history, label):
    rows = []
    for h in history:
        rows.append({
            'family': label,
            'step': h['step'],
            'fixed_count': h['fixed_count'],
            'AIC': h['aic'],
            'LjungBox_p_lag12': h['lb_pvalue_lag12'],
            'fixed_params': ', '.join(sorted(h['fixed'].keys())) if h['fixed
        })
    return pd.DataFrame(rows)
```

Part 1

In [107…

```
ar_hist = simplify_by_zeroing(vw_series, p=9, q=0)
ar_tbl = history_table(ar_hist, 'AR from AR(9)')
display(ar_tbl)
```

```
best_ar = min(ar_hist, key=lambda h: h['aic'])
best_ar_res = best_ar['result']
print('Best AR candidate from simplification path:')
print(f"Step={best_ar['step']}, AIC={best_ar['aic']:.3f}, Ljung-Box p(12)={b
print('Fixed coefficients:', best_ar['fixed'] if best_ar['fixed'] else '(nor
display(best_ar_res.summary())
```

	family	step	fixed_count	AIC	LjungBox_p_lag12	fixed_params
0	AR from AR(9)	0	0	-2950.147877	0.527807	(none)
1	AR from AR(9)	1	1	-2952.134850	0.712816	ar.L2
2	AR from AR(9)	2	2	-2953.452455	0.678840	ar.L2, ar.L6
3	AR from AR(9)	3	3	-2953.888333	0.651856	ar.L2, ar.L6, ar.L7
4	AR from AR(9)	4	4	-2954.624775	0.703580	ar.L2, ar.L4, ar.L6, ar.L7

```
Best AR candidate from simplification path:
Step=4, AIC=-2954.625, Ljung-Box p(12)=0.7036
Fixed coefficients: {'ar.L2': 0.0, 'ar.L6': 0.0, 'ar.L7': 0.0, 'ar.L4': 0.0}
```

SARIMAX Results

Dep. Variable:	vwrtn	**No. Observations:**	996
Model:	SARIMAX(9, 0, 0)	**Log Likelihood**	1483.312
Date:	Tue, 07 Apr 2026	**AIC**	-2954.625
Time:	15:41:09	**BIC**	-2925.257
Sample:	0	**HQIC**	-2943.456
	- 996		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
ar.L1	0.1263	0.019	6.804	0.000	0.090	0.163
ar.L2 (fixed)	0	nan	nan	nan	nan	nan
ar.L3	-0.0867	0.022	-3.912	0.000	-0.130	-0.043
ar.L4 (fixed)	0	nan	nan	nan	nan	nan
ar.L5	0.0958	0.024	3.957	0.000	0.048	0.143
ar.L6 (fixed)	0	nan	nan	nan	nan	nan
ar.L7 (fixed)	0	nan	nan	nan	nan	nan
ar.L8	0.0684	0.021	3.253	0.001	0.027	0.110
ar.L9	0.0898	0.021	4.189	0.000	0.048	0.132
sigma2	0.0029	7.72e-05	37.562	0.000	0.003	0.003

Ljung-Box (L1) (Q):	0.12	**Jarque-Bera (JB):**	1464.56
Prob(Q):	0.73	**Prob(JB):**	0.00
Heteroskedasticity (H):	0.42	**Skew:**	0.11
Prob(H) (two-sided):	0.00	**Kurtosis:**	8.96

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Part 2

In [108…

```
ma_hist = simplify_by_zeroing(vw_series, p=0, q=9)
ma_tbl = history_table(ma_hist, 'MA from MA(9)')
display(ma_tbl)

best_ma = min(ma_hist, key=lambda h: h['aic'])
best_ma_res = best_ma['result']
```

```
print('Best MA candidate from simplification path:')
print(f"Step={best_ma['step']}, AIC={best_ma['aic']:.3f}, Ljung-Box p(12)={b
print('Fixed coefficients:', best_ma['fixed'] if best_ma['fixed'] else '(nor
display(best_ma_res.summary())
```

	family	step	fixed_count	AIC	LjungBox_p_lag12	fixed_params
0	MA from MA(9)	0	0	-2946.106103	0.466140	(none)
1	MA from MA(9)	1	1	-2948.064053	0.654532	ma.L6
2	MA from MA(9)	2	2	-2949.767164	0.829640	ma.L4, ma.L6
3	MA from MA(9)	3	3	-2951.397842	0.915662	ma.L2, ma.L4, ma.L6
4	MA from MA(9)	4	4	-2951.802761	0.926960	ma.L2, ma.L4, ma.L6, ma.L7

```
Best MA candidate from simplification path:
Step=4, AIC=-2951.803, Ljung-Box p(12)=0.9270
Fixed coefficients: {'ma.L6': 0.0, 'ma.L4': 0.0, 'ma.L2': 0.0, 'ma.L7': 0.0}
```

SARIMAX Results

Dep. Variable:	vwrtn	**No. Observations:**	996
Model:	SARIMAX(0, 0, 9)	**Log Likelihood**	1481.901
Date:	Tue, 07 Apr 2026	**AIC**	-2951.803
Time:	15:41:11	**BIC**	-2922.441
Sample:	0	**HQIC**	-2940.636
	- 996		
Covariance Type:	opg		

	coef	**std err**	**z**	**P>|z|**	**[0.025**	**0.975]**
ma.L1	0.1290	0.020	6.341	0.000	0.089	0.169
ma.L2 (fixed)	0	nan	nan	nan	nan	nan
ma.L3	-0.1025	0.023	-4.393	0.000	-0.148	-0.057
ma.L4 (fixed)	0	nan	nan	nan	nan	nan
ma.L5	0.1039	0.025	4.204	0.000	0.055	0.152
ma.L6 (fixed)	0	nan	nan	nan	nan	nan
ma.L7 (fixed)	0	nan	nan	nan	nan	nan
ma.L8	0.0414	0.018	2.263	0.024	0.006	0.077
ma.L9	0.1003	0.020	5.086	0.000	0.062	0.139
sigma2	0.0029	7.63e-05	37.943	0.000	0.003	0.003

Ljung-Box (L1) (Q):	0.17	**Jarque-Bera (JB):**	1469.34
Prob(Q):	0.68	**Prob(JB):**	0.00
Heteroskedasticity (H):	0.43	**Skew:**	0.10
Prob(H) (two-sided):	0.00	**Kurtosis:**	8.98

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Part 3

In [109…

```
arma_hist = simplify_by_zeroing(vw_series, p=6, q=6)
arma_tbl = history_table(arma_hist, 'ARMA from ARMA(6,6)')
display(arma_tbl)

best_arma = min(arma_hist, key=lambda h: h['aic'])
best_arma_res = best_arma['result']
```

```
print('Best ARMA candidate from simplification path:')
print(f"Step={best_arma['step']}, AIC={best_arma['aic']:.3f}, Ljung-Box p(12
print('Fixed coefficients:', best_arma['fixed'] if best_arma['fixed'] else '
display(best_arma_res.summary())
```

	family	step	fixed_count	AIC	LjungBox_p_lag12	fixed_params
0	ARMA from ARMA(6,6)	0	0	-2953.840207	NaN	(none)
1	ARMA from ARMA(6,6)	1	1	-2955.779605	0.038681	ar.L4
2	ARMA from ARMA(6,6)	2	2	-2955.371960	0.030373	ar.L4, ma.L4
3	ARMA from ARMA(6,6)	3	3	-2952.004172	0.058844	ar.L2, ar.L4, ma.L4
4	ARMA from ARMA(6,6)	4	4	-2954.045379	0.123193	ar.L2, ar.L4, ar.L6, ma.L4
5	ARMA from ARMA(6,6)	5	5	-2955.157829	0.179647	ar.L2, ar.L4, ar.L6, ma.L4, ma.L6
6	ARMA from ARMA(6,6)	6	6	-2963.366503	0.145377	ar.L2, ar.L4, ar.L5, ar.L6, ma.L4, ma.L6
7	ARMA from ARMA(6,6)	7	7	-2956.780969	0.193279	ar.L2, ar.L4, ar.L5, ar.L6, ma.L2, ma.L4, ma.L6

```
Best ARMA candidate from simplification path:
Step=6, AIC=-2963.367, Ljung-Box p(12)=0.1454
Fixed coefficients: {'ar.L4': 0.0, 'ma.L4': 0.0, 'ar.L2': 0.0, 'ar.L6': 0.0,
'ma.L6': 0.0, 'ar.L5': 0.0}
```

SARIMAX Results

Dep. Variable:	vwrtn	**No. Observations:**	996
Model:	SARIMAX(6, 0, 6)	**Log Likelihood**	1488.683
Date:	Tue, 07 Apr 2026	**AIC**	-2963.367
Time:	15:41:19	**BIC**	-2929.090
Sample:	0	**HQIC**	-2950.332
	- 996		
Covariance Type:	opg		

	coef	**std err**	**z**	**P>|z|**	**[0.025**	**0.975]**
ar.L1	0.5732	0.092	6.242	0.000	0.393	0.753
ar.L2 (fixed)	0	nan	nan	nan	nan	nan
ar.L3	0.4261	0.093	4.599	0.000	0.245	0.608
ar.L4 (fixed)	0	nan	nan	nan	nan	nan
ar.L5 (fixed)	0	nan	nan	nan	nan	nan
ar.L6 (fixed)	0	nan	nan	nan	nan	nan
ma.L1	-0.4557	0.438	-1.041	0.298	-1.314	0.402
ma.L2	-0.0608	0.264	-0.231	0.818	-0.577	0.456
ma.L3	-0.5458	0.302	-1.805	0.071	-1.138	0.047
ma.L4 (fixed)	0	nan	nan	nan	nan	nan
ma.L5	0.0623	0.042	1.477	0.140	-0.020	0.145
ma.L6 (fixed)	0	nan	nan	nan	nan	nan
sigma2	0.0029	0.001	2.002	0.045	6.05e-05	0.006

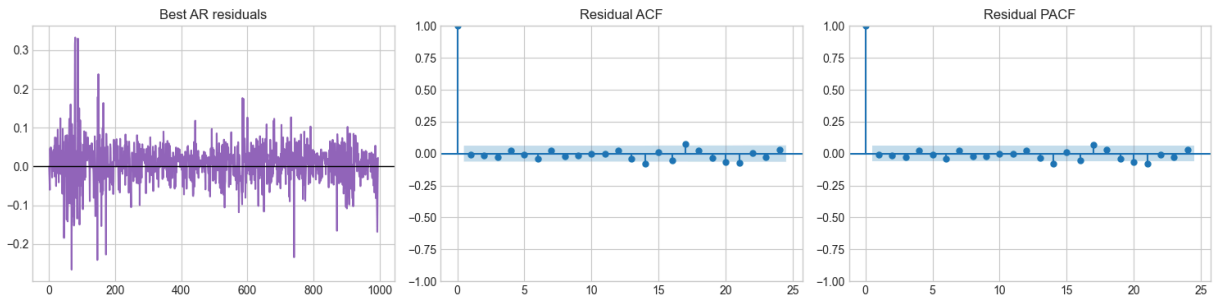
Ljung-Box (L1) (Q):	0.00	**Jarque-Bera (JB):**	1829.18
Prob(Q):	0.95	**Prob(JB):**	0.00
Heteroskedasticity (H):	0.41	**Skew:**	0.16
Prob(H) (two-sided):	0.00	**Kurtosis:**	9.65

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

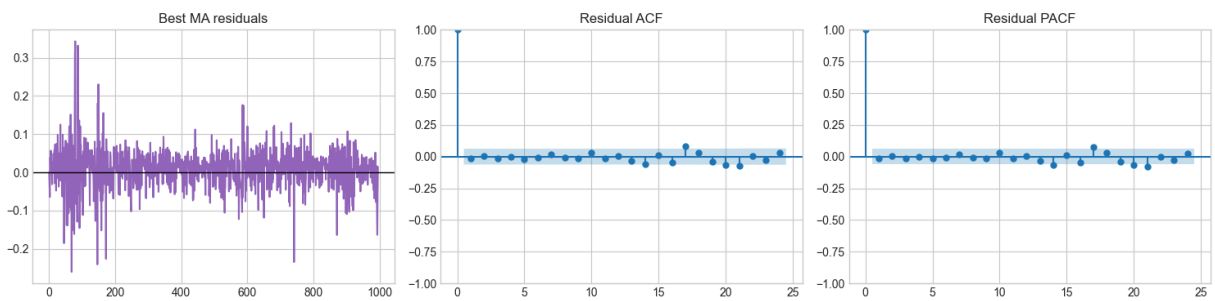
In [110…

```
def residual_diagnostics(res, title):
    resid = pd.Series(res.resid).dropna()
    fig, axes = plt.subplots(1, 3, figsize=(15, 3.8))
    axes[0].plot(resid.values, color='tab:purple')
```

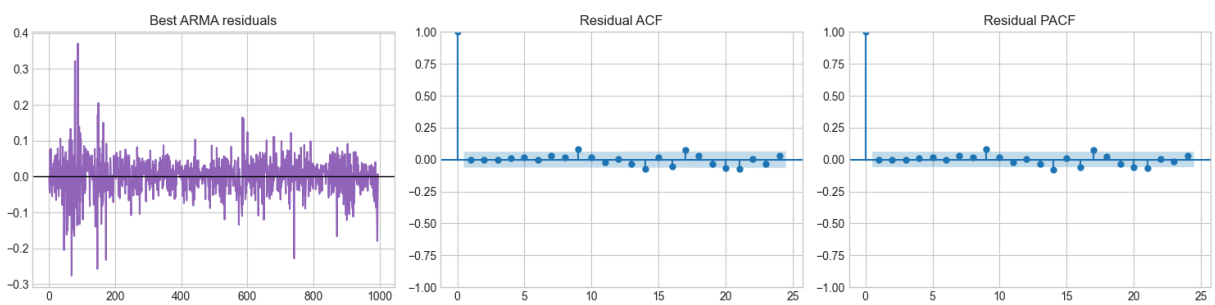
```
    axes[0].axhline(0, color='black', linewidth=1)
    axes[0].set_title(f'{title} residuals')
    plot_acf(resid, lags=24, ax=axes[1])
    axes[1].set_title('Residual ACF')
    plot_pacf(resid, lags=24, ax=axes[2], method='ywm')
    axes[2].set_title('Residual PACF')
    plt.tight_layout()
    plt.show()

    print(acorr_ljungbox(resid, lags=[12], return_df=True))

residual_diagnostics(best_ar_res, 'Best AR')
residual_diagnostics(best_ma_res, 'Best MA')
residual_diagnostics(best_arma_res, 'Best ARMA')
```




```
     lb_stat  lb_pvalue
12   4.64181   0.968886
```




```
     lb_stat  lb_pvalue
12   2.501815   0.998155
```




```
     lb_stat  lb_pvalue
12   9.540642   0.656189
```

From Lecture 05, hardcoding the models we covered:

vw.ar9.1: fixed=c(NA,0,NA,0,NA,0,0,NA,NA,NA)

vw.ma9.3: fixed=c(NA,0,NA,0,NA,0,0,0,NA,NA)

vw.66.1: fixed=c(NA,0,NA,NA,0,NA,NA,0,NA,NA,NA,NA,NA)

In [113…

```
lecture_targets = pd.DataFrame([
    {'lecture_model':'vw.ar9.1','AIC_lecture':-3000.226,'Ljung_p_lecture':0.
    {'lecture_model':'vw.ma9.3','AIC_lecture':-3005.032,'Ljung_p_lecture':0.
    {'lecture_model':'vw.66.1','AIC_lecture':-3013.093,'Ljung_p_lecture':0.6
])


bench_specs = {
    'vw.ar9.1': {'p':9, 'q':0, 'fixed': {'ar.L2':0.0, 'ar.L4':0.0, 'ar.L6':0
    'vw.ma9.3': {'p':0, 'q':9, 'fixed': {'ma.L2':0.0, 'ma.L4':0.0, 'ma.L6':0
    'vw.66.1': {'p':6, 'q':6, 'fixed': {'ar.L2':0.0, 'ar.L5':0.0, 'ma.L2':0.
}

bench_rows = []
for name, spec in bench_specs.items():
    model = SARIMAX(vw_series, order=(spec['p'], 0, spec['q']), trend='c',
                    enforce_stationarity=False, enforce_invertibility=False)
    with model.fix_params(spec['fixed']):
        fit = model.fit(disp=False)

    lb = acorr_ljungbox(fit.resid, lags=[18], model_df=(spec['p'] + spec['q'
    bench_rows.append({
        'lecture_model': name,
        'AIC_reproduced': float(fit.aic),
        'Ljung_p_reproduced_lag18': float(lb['lb_pvalue'].iloc[0]),
        'fixed_params': ', '.join(sorted(spec['fixed'].keys()))
    })

bench_df = pd.DataFrame(bench_rows)
compare_lecture = lecture_targets.merge(bench_df, on='lecture_model', how='l
compare_lecture['AIC_diff_reproduced_minus_lecture'] = compare_lecture['AIC_
compare_lecture['Ljung_diff_reproduced_minus_lecture'] = compare_lecture['Lj
display(compare_lecture)

my_best = pd.DataFrame([
    {'model':'Best AR','AIC_my':best_ar['aic']},
    {'model':'Best MA','AIC_my':best_ma['aic']},
    {'model':'Best ARMA','AIC_my':best_arma['aic']},
])
display(my_best)
```

	lecture_model	AIC_lecture	Ljung_p_lecture	AIC_reproduced	Ljung_p_reproduced_l
0	vw.ar9.1	-3000.226	0.3667	-2967.738436	0.11
1	vw.ma9.3	-3005.032	0.5157	-2969.138383	0.25
2	vw.66.1	-3013.093	0.6979	-2963.374960	0.00

	model	AIC_my
0	Best AR	-2954.624775
1	Best MA	-2951.802761
2	Best ARMA	-2963.366503

In [114…

```
compare = pd.DataFrame([
    {'model':'Best AR (from AR9 path)','AIC':best_ar['aic'],'LjungBox_p_lag1
    {'model':'Best MA (from MA9 path)','AIC':best_ma['aic'],'LjungBox_p_lag1
    {'model':'Best ARMA (from 6,6 path)','AIC':best_arma['aic'],'LjungBox_p_
]).sort_values('AIC').reset_index(drop=True)
display(compare)
```

	model	AIC	LjungBox_p_lag12	fixed
0	Best ARMA (from 6,6 path)	-2963.366503	0.145377	{'ar.L4': 0.0, 'ma.L4': 0.0, 'ar.L2': 0.0, 'ar...
1	Best AR (from AR9 path)	-2954.624775	0.703580	{'ar.L2': 0.0, 'ar.L6': 0.0, 'ar.L7': 0.0, 'ar...
2	Best MA (from MA9 path)	-2951.802761	0.926960	{'ma.L6': 0.0, 'ma.L4': 0.0, 'ma.L2': 0.0, 'ma...

For Problem 2, I started with AR(9), MA(9), and ARMA(6,6), and then simplified each model step by step by setting the least significant coefficient to zero and refitting. After each change, I checked AIC and residual diagnostics (Ljung-Box plus residual ACF/PACF) to make sure the model still looked reasonable. I also compared my simplified models to the lecture versions (`vw.ar9.1` , `vw.ma9.3` , and `vw.66.1`) -- hardcoded those based on the lecture html from Canvas

My main finding is that simplification helped across all three parts: removing weak terms kept residuals acceptable, and generally got better AIC than the unsimplified starting models. The strongest overall result came from the ARMA(6,6)-based simplification, which matches the lecture takeaway that the constrained ARMA model is the best of the three. My exact numbers are a little different from lecture because I used Python/statsmodels instead of R, but the overall conclusion is the same.